

When Does a Scientific Calculation Apply?

Condition-Aware Verification of LLM-Generated Scientific Claims

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Program Officer: Dr. Richard D. (Doug) Riecken

Proposer: Xinya Du

Institution: The University of Texas at Dallas

Email: Xinya.Du@UTDallas.edu

Phone: +1 (972) 883-2634

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Proposed Research and Motivation

A calculation can be correct while the scientific conclusion it supports is wrong. Suppose a language model reads that gas in a sealed, rigid vessel is heated from 300 K to 600 K and concludes that its volume doubles. A calculator confirms the temperature ratio of two. Yet the volume cannot change in a rigid vessel: under the ideal-gas model with fixed gas amount, pressure doubles instead. The volume-temperature relation would require constant pressure [11]. The failure is not arithmetic. It is a failure to establish which scientific relation applies to the described situation. Harder cases place the deciding condition in another passage or establish it only through an earlier check; Task 2 varies these dependencies.

This project asks: *How can a system establish that the conditions needed for a scientific calculation hold, and preserve those conditions as results pass between different reasoning methods?* We will develop a verifier that connects evidence in scientific text to quantitative calculations and logical checks. Each component will return both its result and the conditions under which that result is justified. The system will identify unsupported conditions, seek relevant evidence, and revise a claim only when the revised reasoning is supported.

Our hypothesis is that making these conditions explicit and checking them across component boundaries will reduce false acceptance of scientific claims compared with tool execution plus a learned correctness score. The benefit may disappear if extracting conditions introduces more errors than it detects, or if conservative checks reject too many valid claims. We will test both possibilities at matched computational cost and coverage. The initial scope is quantitative claims about elementary thermodynamic and mechanical systems, expressed in short scientific passages and tables. SciBench identifies assumption identification as a source of scientific reasoning errors, and MathTrap finds failures to combine known facts with conditions that invalidate standard solutions [16, 19]. These earlier findings motivate testing contemporary models; they do not establish their error rates.

The project connects to a longstanding architectural idea: intelligence requires coordination among different representations and reasoning methods. Riecken’s agent architecture combined spatial, structural, functional, temporal, and causal reasoning [14]; McCarthy, Minsky, Riecken, and colleagues further argued that each technique must determine whether it suits a task [10]. Our proposed contribution is a way to make the conditions for using and combining these methods explicit and testable when those conditions must be recovered from scientific sources.

Relation to current work. Classical compositional modeling represented operating assumptions and checked model behavior against them using hand-encoded domain theories [5]. FacTool and AutoVerifier organize tool-based checking and technical evidence [1, 4]; ProgramFC decomposes claims into programs of specialized checks [13]. Logic-LM and SatLM connect language-generated specifications to symbolic solvers [12, 18]. Neuro-symbolic PRM already distinguishes symbolic validity from semantic grounding and scores verifier-passing steps [20]. Our question is how to recover conditions from scientific text, measure omitted conditions and invented restrictions, and preserve conditions across compo-

nent boundaries. A companion CCMI white paper studies uncertain document evidence and argument revision; this project studies scientific principles, their applicability conditions, units, and variable bindings across calculation and constraint solving.

Research Hypothesis and Technical Approach

The three tasks share one object: a scientific claim linked to the evidence, principles, and conditions needed to justify it. Task 1 constructs that representation; Task 2 checks and combines its parts; Task 3 uses specific failures to guide correction.

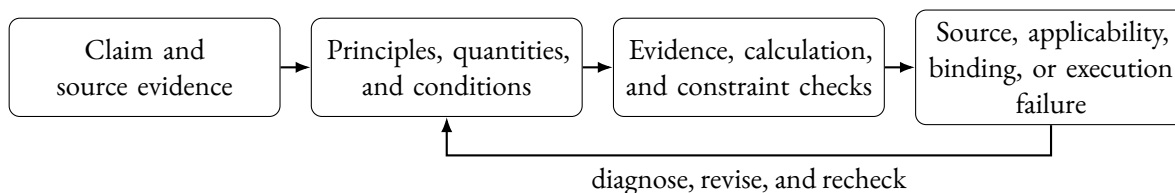


Figure 1: Checks preserve conditions and diagnose source, applicability, binding, or execution failures to guide retrieval and correction.

Task 1: Recover Scientific Principles and Their Conditions

A useful representation must preserve more than an equation. We will construct a compact record containing the principle and source passage; variables, units, entities, and measurement conditions; the executable relation; and its applicability conditions. Observed facts, modeling assumptions, and derived results will be distinguished. In the vessel example, “sealed” supports fixed gas amount and “rigid” supports fixed volume; neither supports constant pressure. The ideal-gas approximation must also be stated or justified by the supplied context.

Starting from a claim, the system will retrieve a small set of relevant scientific sources and extract only the principles needed for verification. A language model will propose the records. Type and unit checks will detect structural errors, while evidence checks will assess whether the extracted conditions and quantity bindings preserve the source’s meaning. A missing condition will remain unknown rather than be silently assumed true. Conflicting sources will retain their separate conditions instead of being merged into one apparently supported rule.

This builds on the PI’s document-level template filling [3] and co-authored scientific event extraction benchmark SciEvent [2]. The new challenge is recovering the restrictions under which extracted knowledge can be used. We will compare extracting equations alone, extracting equations with nearby qualifiers, and retrieving additional context to resolve missing conditions. Independent annotations will measure both omitted conditions and invented restrictions. Fully specified records supplied by annotators will isolate failures of extraction from failures of subsequent reasoning.

Research question. Which source information is necessary for a reusable principle record, and when does recovering additional context improve verification enough to justify its cost?

Task 2: Coordinate Checks Without Losing Their Assumptions

We will develop a planner that turns the claim into a finite set of required checks. Three components will share the principle records: an evidence checker links facts and conditions to passages; a calculation engine evaluates equations and units; and a constraint solver checks compatibility among conditions. The initial logical backend will use satisfiability modulo theories (SMT), supporting explicit numerical and logical constraints. This builds on the PI’s work on logical interpretation of natural-language explanations [6].

The planner will work backward from the target quantity or relation. It will retrieve candidate principles, identify their required inputs and conditions, and assign each requirement to a component. Checks that use another component’s output will also inherit the conditions attached to that output. For example, a calculation of doubled volume cannot be accepted merely because its arithmetic passes: the evidence checker must establish constant pressure, and the constraint checker must reconcile that requirement with the rigid-vessel description. Unresolved requirements trigger targeted retrieval or an unresolved verdict; they cannot be satisfied by the claim under examination itself.

Each component will return a *verification receipt*: its principle, source, variable bindings, operation, result, and supported, contradicted, or unresolved conditions. The central distinction is between **execution validity** and **semantic groundedness**. A successful execution will not itself be treated as proof of scientific correctness; semantic applicability remains an explicit check. Evidence judgments remain fallible even when calculations and encoded constraints are checked exactly.

The central experiment will test whether explicit exchange of conditions improves decisions beyond giving the same evidence and tools to a neural planner or a verifier-first process reward model. We will vary whether a relevant condition appears beside its equation, in another passage, or only as the conclusion of an earlier check. This tests a specific difficulty of composition: individually acceptable intermediate results may depend on incompatible operating conditions or refer to different objects or times.

For finite plans, we will specify acceptance rules that preserve required conditions and prevent unresolved conditions from becoming supported during composition. Conditional correctness is a design requirement; the research examines which source and context information must accompany intermediate results and how interface failures affect end-to-end false acceptance. We will analyze restricted compositions and test the limits experimentally, without promising formal guarantees for unrestricted scientific text.

Research question. What information must specialized checkers exchange so that locally valid results support a coherent scientific conclusion?

Task 3: Diagnose and Correct the Failed Scientific Inference

Verification receipts will distinguish errors in source support, principle applicability, quantity binding, and execution. A model will propose a correction using the failed requirement and its evidence. In the vessel example, the corrected statement says that volume remains fixed; pressure doubles under the stated ideal-gas and fixed-amount assumptions. If those assumptions cannot be established, the output will explain what remains unresolved rather than invent evidence.

A revised claim will be accepted only if the failed checks are resolved without invalidating components that previously passed. The system will recheck the revised claim and any dependent results. It must preserve the original question, objects, and operating setting; deleting a difficult condition or replacing a specific assertion with a vague one will not count as repair. Corrections that legitimately narrow a claim’s scope will be labeled separately from corrections within its original scope.

The PI’s FaithScore decomposes outputs for factual assessment, and FG-PRM detects fine-grained errors in mathematical reasoning [7, 8]. Extending such diagnoses to applicability and binding errors is proposed work. We will first use prompted correction with external feedback, comparing typed receipts with solver-error feedback and full re-verification at matched cost. Training on verified counterfactual examples will be optional, pursued only if the initial system shows a clear need. Such examples will not supply the final evaluation labels.

Research question. Does identifying the failed scientific requirement improve correction while preserving the claim’s intended scope and avoiding regressions?

Evaluation and Measures of Success

The first comparison uses paired cases that change one applicability condition while preserving the numerical calculation. Evaluation will combine controlled cases with document-based claims. Annotators with relevant scientific training will independently construct cases from public scientific sources, including held-out principles and compositions, with valid claims and matched applicability or binding violations. The source documents will be available to every system; reference encodings and labels will remain hidden. Splits will separate principles and source documents, not merely paraphrases. Human-authored and automatically perturbed results will be reported separately. SciFact and SCITAB provide complementary tests of textual evidence and numerical table reasoning [9, 15]; neither alone measures the proposed applicability problem.

The primary outcome is the fraction of accepted claims that are invalid, at matched coverage (the fraction accepted) and total cost, including retrieval, model calls, execution, and retries. We will also report rejection of valid claims, unresolved cases, condition-extraction accuracy, error localization, and scope-preserving correction without regression. Labels will come from independent evidence and step annotations, not the system’s own judgments. Cases without sufficient information will have an unresolved reference outcome. This prevents higher abstention from masquerading as better verification.

Baselines will include retrieval plus an LLM judge, a ProgramFC-style planner adapted to the same scientific tools without explicit condition exchange [13], verifier-first process scoring, and symbolic checking supplied with human-validated records. Ablations will remove condition extraction, condition exchange between components, or typed repair feedback. The primary benchmark is a lower rate of invalid accepted claims than the strongest tool-augmented baseline at matched coverage and cost on held-out principles, with the target margin and sample size fixed before testing. We will report uncertainty intervals and failures from extraction or overly restrictive checking.

Potential Impact on DAF and DoW Capabilities

The Science of Information, Computation, Learning, and Fusion portfolio calls for connecting data-driven extraction with knowledge-driven reasoning and integrating diverse reasoning components. The proposed work addresses these objectives by studying what makes their combined conclusions defensible.

Motivating settings include reviewing AI-generated engineering analyses and checking whether measurements support claims under documented operating conditions. An analyst would receive the relevant source, calculation, conditions, and a specific explanation of any unsupported step. This could reduce repeated manual checking and make incorrect uses of otherwise valid principles easier to detect. The project will use public sources and unclassified document tasks; it requires neither an operational deployment nor access to restricted mission data.

Team and Development Plan

PI Xinya Du will lead the work at UT Dallas with graduate and undergraduate researchers. The PI's research in scientific information extraction, logical reasoning, fine-grained evaluation, and scientific hypothesis discovery provides the foundation [2, 6, 8, 17]. No external collaborators or subawards are planned. The estimated three-year budget is \$450,000 including indirect costs, supporting personnel, computation, independent annotation, and dissemination.

Year 1 will deliver an end-to-end prototype evaluated on independently annotated thermodynamic cases. **Year 2** will study condition extraction and composition, using controlled failures to refine the interfaces and acceptance rules. **Year 3** will evaluate correction and transfer to held-out mechanical relations, retaining the same core representation. Deliverables include annotated test cases, principle and receipt specifications, reproducible implementations, and research publications. Expansion to additional domains or repair-model training will depend on the results of the core studies.

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